

COLLISION-MARKED TRAJECTORY CLUSTERS AND JOINT DYNAMIC LARGE DEVIATIONS FOR DETERMINISTIC HARD SPHERES

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ABSTRACT. Every actual deterministic hard-sphere contact is marked on a trace-regular correlation hierarchy. Incoming flux traces, rather than interior mass alone, control future contact exponential moments and exclude boundary-layer counterexamples. A genealogical controllability Gramian containing root translations, root velocities, and all creation variables gives a first-surplus codimension-two tube estimate. The true reflected future is retained and propagated by the trace semigroup. Fixed-horizon source sewing, conservative action-dense regularization, and normalized deterministic tilts establish grand-canonical and micro-canonical joint density–contact LDPs.

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1. ROUND-SIX POSITIVE CLOSURE: TRACE-REGULAR ACTUAL CONTACTS AND THE JOINT DYNAMIC LDP

The grand-canonical construction is logically prior to B1. Collision exponential moments are proved only on the actual initial and tilted hierarchy class, which carries an incoming-boundary trace norm. This excludes the boundary-layer counterexample for arbitrary interior L^1 densities. The first surplus contact is controlled by a full genealogical Gramian containing root translations as well as creation variables; after that contact the true post-collisional future is retained and propagated by the trace-class semigroup.

1.1. Orientation and exact balance. For $i < j$ at contact, put

$$\omega = \frac{x_j - x_i}{\varepsilon}, \quad (v_i - v_j) \cdot \omega > 0$$

for the incoming orientation, and use the elastic reflection

$$v'_i = v_i - ((v_i - v_j) \cdot \omega)\omega, \quad v'_j = v_j + ((v_i - v_j) \cdot \omega)\omega.$$

Every unordered contact is counted once in Γ_T^ε , on the quotient $(v, v_*, \omega) \sim (v_*, v, -\omega)$.

Proposition 1.1 (Microscopic density–contact balance). *For every weighted smooth test p ,*

$$\langle p_T, \pi_T^\varepsilon \rangle - \langle p_0, \pi_0^\varepsilon \rangle - \int_0^T \langle (\partial_t + v \cdot \nabla_x)p, \pi_t^\varepsilon \rangle dt = \int \Delta_\omega p d\Gamma_T^\varepsilon.$$

The identity is invariant under the exchange quotient.

Proof. Integrate each particle test on free-flight intervals. Interior endpoints telescope. The jump of the two colliding particle tests is exactly $\Delta_\omega p$. Grazing and simultaneous multiple contacts form a Liouville-null set. Exchanging the labels and replacing ω by $-\omega$ leaves the unordered jump unchanged. $\square$

1.2. A trace-regular hierarchy class. Let g_k be the rescaled k -particle correlation density on the hard-core domain D_k^ε . On an incoming face $\partial_{ij}^- D_k^\varepsilon$, define the flux trace

$$\mathsf{T}_{ij}^- g_k = ((v_i - v_j) \cdot \omega)_+ \mathsf{Tr}_{ij}^- g_k.$$

For $\beta > 0$ and label radius α , put

$$\begin{aligned} \|G\|_{\alpha, \beta}^{\text{tr}} &= \sum_{k \geq 0} \frac{e^{\alpha k}}{k!} \int_{D_k^\varepsilon} |g_k| e^{\beta |V_k|^2} dZ_k \\ &\quad + \sum_{k \geq 2} \frac{e^{\alpha k}}{k!} \sum_{i < j} \int_{\partial_{ij}^- D_k^\varepsilon} \left(|\mathsf{T}_{ij}^- g_k| + \tau_* |\nabla_{\tan} \mathsf{T}_{ij}^- g_k| \right) e^{\beta |V_k|^2} d\sigma. \end{aligned}$$

The tangential term is needed only on the regular contact charts; grazing pieces are placed in the weighted bad-set ledger.

Lemma 1.2 (Initial and tilted trace regularity). *The hard-core grand-canonical Maxwellian hierarchy belongs to $\mathfrak{X}_{\alpha,\beta}^{\text{tr}}$ for every finite α and every β below the Maxwellian exponent. For a bounded particle/contact source of size R , the exact Feynman–Kac BBGKY evolution maps*

$$\mathfrak{X}_{\alpha,\beta}^{\text{tr}} \longrightarrow \mathfrak{X}_{\alpha-Cte^{CR},\beta'}^{\text{tr}}$$

for $0 \leq t \leq T$, with $0 < \beta' < \beta$, and

$$\|\mathcal{U}_{\varepsilon,R}(t)G\|_{\alpha-Cte^{CR},\beta'}^{\text{tr}} \leq e^{Cte^{CR}} \|G\|_{\alpha,\beta}^{\text{tr}}.$$

The constants are uniform in ε .

Proof. The Maxwellian hierarchy has factorial correlations bounded by $C^k e^{-\beta_0 |V_k|^2}$. The hard-core boundary trace is its one-sided restriction multiplied by the incoming flux, which is integrable after reducing the Gaussian exponent. Tangential derivatives obey the same bound.

For the evolution, integrate the BBGKY equations on each hard-core domain. Free transport preserves the interior weighted mass. The Green identity pairs the outgoing and incoming traces through the elastic reflection, whose velocity Jacobian and energy are one. A contact source multiplies an incoming trace by at most e^R ; a particle source costs the same factor per created label. Summing the boundary operators with the factorial weight loses Cte^{CR} in the label radius and a fixed amount in the Gaussian exponent. The tangential trace equation has the same triangular form plus bounded curvature coefficients on regular charts. Gronwall's inequality gives the estimate. $\square$

Corollary 1.3 (Correct contact exponential ledger). *For $G \in \mathfrak{X}_{\alpha,\beta}^{\text{tr}}$,*

$$\mathbb{E}_G e^{zN_t} \leq \exp\{Cte^{Cz}\} \|G\|_{\alpha,\beta}^{\text{tr}}$$

after normalization by the initial mass, uniformly for bounded z . The same estimate holds after a stopping contact for the reflected outgoing trace state. No corresponding estimate is claimed for arbitrary interior L^1 densities without trace control.

Proof. Insert the constant contact source z in Lemma 1.2. At a stopping contact, elastic reflection maps the incoming trace isometrically to an outgoing trace, which is an admissible initial boundary coordinate for the remaining semigroup. The trace norm, rather than the interior mass alone, controls the next contact. $\square$

1.3. The full genealogical controllability Gramian. Fix a creation forest and its regular chronological cell. For a proposed first surplus contact between labels a, b , let $\mathcal{P}(a, b)$ be the union of their two exclusive ancestral paths up to their first common ancestor. The control variables are:

- (i) the three root-position translations of each independent root component;

(ii) every creation time and the two local impact-normal coordinates on $\mathcal{P}(a, b)$;

(iii) the root velocity coordinates not fixed by the energy cutoff.

Let $J_e(s)$ be the derivative of the relative position $x_a(s) - x_b(s)$ with respect to control e , transported through every actual reflection. With $\Pi_{ab}^\perp$ the projection onto the plane orthogonal to the relative velocity, define

$$\mathcal{G}_{ab}(s) = \sum_{e \in \mathcal{U}(a, b)} \Pi_{ab}^\perp J_e(s) J_e(s)^T \Pi_{ab}^\perp.$$

Lemma 1.4 (Quantitative genealogical observability). *On the regular set where all relative speeds are at least δ , all incidence cosines are at least δ , and chronological gaps are at least δ ,*

$$\lambda_{\min}(\mathcal{G}_{ab}(s)|_{(v_a - v_b)^\perp}) \geq c \delta^{C(1+m)} (1+m)^{-C},$$

where $m = |\mathcal{P}(a, b)|$. *The complement of the regular set has Gaussian-weighted activity at most*

$$C^m (m\delta + \delta^c).$$

The constants do not rely on a uniform analytic-sublevel exponent.

Proof. Write the linearized free flight as $(\delta x, \delta v) \mapsto (\delta x + t\delta v, \delta v)$. At a non-grazing reflection, the linearized scattering matrix and its inverse have norm at most $C\delta^{-C}$. Suppose a transverse covector ℓ annihilates every column of the control matrix. Propagate ℓ backward through the inverse scattering matrices. Orthogonality to the two impact normal controls at each creation edge removes the two transverse components introduced there. At the terminal roots, orthogonality to the explicitly included independent root translations removes the remaining transverse components. Thus $\ell = 0$, proving full rank.

For the quantitative estimate, normalize $|\ell| = 1$ and iterate the inverse matrix bounds. At some control column the pairing is at least $c\delta^{C(1+m)}(1+m)^{-C}$; otherwise the backward recursion would leave a nonzero root component while all root translations are annihilated. Squaring and summing gives the Gramian bound. Small-speed and grazing sets have direct Gaussian/angular measure $O(\delta^c)$; a gap shorter than δ has time-simplex measure $O(m\delta)$. A union bound over the chronological cell gives the bad-set estimate, with no appeal to orders of vanishing of composed analytic minors. $\square$

Lemma 1.5 (First-surplus contact tube). *Condition on every variable before the first surplus contact and on all controls except a two-dimensional subfamily selected by Lemma 1.4. The set of controls producing the contact has relative activity at most*

$$C^m (1 + \mathbb{V})^C (1 + T)^C \varepsilon^\alpha$$

for one $\alpha > 0$, after optimizing $\delta = \varepsilon^\kappa$. *The estimate is unchanged by bounded particle and actual-contact marks.*

Proof. On the regular set, the map from the selected controls to the transverse relative position has a two-dimensional Jacobian bounded below by the Gramian estimate. Coarea bounds the preimage of the contact disk of radius ε by $C^m \varepsilon^2 \delta^{-C(1+m)}$. The chronological tree weight supplies one small factor per additional ancestral edge, so after summing in m the powers of δ^{-Cm} are absorbed by choosing the trace/label radius and then $\delta = \varepsilon^\kappa$ with fixed small κ . The bad set from Lemma 1.4 contributes $C^m(m\delta + \delta^c)$. Taking the smaller resulting power gives $\alpha > 0$. Bounded marks multiply the integrand by a uniform factor and do not change the coarea map. $\square$

1.4. All later contacts on the true future. After the first surplus contact, keep the reflected velocities and every subsequent actual contact. The post-contact state is an outgoing boundary trace, transported by the exact reflected trace semigroup.

Theorem 1.6 (Future-preserving cyclic-sector estimate). *For a connected k -label cluster, the total marked activity of histories containing at least one surplus contact is bounded by*

$$\varepsilon^\alpha k! C^k (T + \varepsilon)^{k-1} \exp\{CTe^{R+U}\},$$

where R bounds the sources and U is the exponential-generating variable for marked actual contacts. Only the first surplus contact supplies the geometric ε^α factor; the complete deterministic future is controlled by Corollary 1.3 and is never altered or deleted.

Proof. Sum the creation forests by collision coarea, giving $k! C^k (T + \varepsilon)^{k-1}$. Select the chronologically first surplus contact and apply Lemma 1.5. Its post-collisional state is retained exactly. Conditional on that trace state, the exponential generating function of every later actual contact is bounded by Corollary 1.3. Chronological integration yields the exponential factor. Summing the ancestral depth is geometric in the hierarchy label radius. No map to a collision-deleted pseudo-trajectory is used. $\square$

1.5. Grand-canonical marked pressure on fixed horizons. Let

$$Q_{\varepsilon, T}^{\text{gc}}(h, \psi) = \frac{1}{\mu_\varepsilon} \log \mathbb{E}_\varepsilon^{\text{gc}} \exp \{ \mu_\varepsilon (\langle h, \Pi_T^\varepsilon \rangle + \langle \psi, \Gamma_T^\varepsilon \rangle) \}.$$

Theorem 1.7 (Actual-contact cluster pressure). *For one fixed short physical horizon $T > 0$ and every bounded real source ball of radius R , the pressure above has a normally convergent real-trajectory expansion, locally uniform with all finite source derivatives. It converges to a differentiable pressure Q_T^{gc} . On a complex neighborhood of every compact real subball the limit is analytic. Every ψ -derivative marks an actual microscopic contact, and cyclic clusters vanish at rate $O(\varepsilon^\alpha)$.*

Proof. For a fixed R , choose the initial label radius $\alpha_R > CTe^{CR}$. The grand-canonical Maxwellian hierarchy belongs to that trace class by Lemma 1.2.

Creation trees are summed by the usual factorial tree estimate, while Theorem 1.6 controls every cyclic history. These bounds are normally summable and remain so after source differentiation.

To keep the physical horizon fixed as R grows, divide it into a finite number N_R of blocks. Exact finite-volume Feynman–Kac composition acts on the complete trace hierarchy. Lemma 1.2 supplies a uniform stability bound on each block and consumes only a finite amount of label radius. For fixed R , N_R is finite; the local Boltzmann–Grad consistency error is uniform on the reached trace ball, so the sum of the block errors tends to zero with ε . This yields the fixed-horizon pressure on every bounded source ball. Cauchy estimates give source derivatives, and an actual-contact derivative follows directly from the microscopic exponent. $\square$

Deleting the last collision of a limiting creation tree splits it into two rooted trees. Therefore the limiting Hamiltonian is

$$\mathbb{H}(f, p, \psi) = \langle f, v \cdot \nabla_x p \rangle + \frac{1}{2} \int f f_* ((v - v_*) \cdot \omega)_+ (e^{\Delta_{\omega} p + \psi} - 1).$$

Normal convergence justifies the marked Hamilton–Jacobi equation and all first and second source derivatives.

1.6. Exponential compactness. Use the state topology generated by weighted bounded-Lipschitz density tests, bounded continuous contact tests, and one Maxwellian Lyapunov moment $e^{\beta'|v|^2}$, $0 < \beta' < \beta$.

Lemma 1.8 (Joint exponential compact containment). *For every M there is a compact set K_M in the density–contact path space such that*

$$\limsup_{\varepsilon \rightarrow 0} \frac{1}{\mu_\varepsilon} \log \mathbb{P}_\varepsilon^{\text{gc}} \{(\pi^\varepsilon, \Gamma^\varepsilon) \notin K_M\} \leq -M.$$

Proof. A sufficiently small positive Maxwellian particle source lies in the B2 source chart and gives exponential control of the Lyapunov moment. A positive constant contact source gives exponential control of total contact mass by Theorem 1.7. The exact balance in Proposition 1.1 controls time increments of a countable dense family of smooth weighted tests. Weighted Prokhorov compactness and the Skorokhod modulus criterion yield K_M . $\square$

1.7. Upper and lower bounds. For a density path f , define

$$dA_f = \frac{1}{2} f f_* ((v - v_*) \cdot \omega)_+ dt dx dv dv_* d\omega.$$

A pair (f, Γ) is feasible when it satisfies the closed weak balance, exchange symmetry, and finite Lyapunov moment.

Lemma 1.9 (Conservative action-dense regularization). *Every finite-action feasible pair can be approximated by smooth pairs (f_h, Γ_h) with*

$$0 < c_h \leq \frac{d\Gamma_h}{dA_{f_h}} \leq C_h < \infty,$$

which satisfy the balance exactly and whose entropy actions converge.

Proof. First truncate velocities and mollify in space, time, and velocity away from endpoints. The resulting balance defect is $O(h^M)$ in a negative Sobolev norm for arbitrarily large M , because the original balance is exact and the mollifier has matching moments. Partition each compact energy–momentum shell into cells of diameter h . The actual binary-collision incidence matrix spans the orthogonal complement of mass, momentum, and energy. Its right inverse has norm at most $Ch^{-\kappa}$, obtained from paths of admissible cell collisions of length $O(h^{-\kappa})$. Solve for a signed collision correction of size $O(h^{M-\kappa})$.

Add beforehand a positive reference collision background of size η_h , with $h^{M-\kappa} = o(\eta_h)$ and $\eta_h \rightarrow 0$. The corrected collision density remains positive. The relative-entropy cost of the background and correction tends to zero by the quadratic expansion of $q \log q - q + 1$ on compact positive intervals, followed by monotone truncation. Endpoint corrections are inserted in vanishing time layers. This proves exact balance and action convergence. $\square$

Theorem 1.10 (Grand-canonical joint actual-contact LDP). *The pair $(\pi^\varepsilon, \Gamma^\varepsilon)$ satisfies a full good LDP with speed μ_ε and rate*

$$I_T^{\text{gc}}(f, \Gamma) = I_0^{\text{gc}}(f_0) + \begin{cases} \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f, & (f, \Gamma) \text{ feasible and } \Gamma \ll A_f, \\ +\infty, & \text{otherwise,} \end{cases}$$

where $\ell(q) = q \log q - q + 1$.

Proof. On each bounded source ball, Theorem 1.7 gives the limiting cumulant. The union of bounded particle/contact cylinder sources is separating. Lemma 1.8 upgrades finite-dimensional upper bounds to the full closed-set upper bound. Dualizing the density source imposes the balance; dualizing the global union of bounded contact sources and then taking monotone truncations gives the integral conjugate ℓ , including infinite cost for singular current.

For the lower bound, first take a smooth positive pair from Lemma 1.9. Solve the backward balance equation for a bounded test p and put $\psi = \log(d\Gamma/dA_f) - \Delta p$. Define the exact normalized finite-volume Feynman–Kac tilt

$$\frac{d\mathbb{P}_\varepsilon^{p, \psi}}{d\mathbb{P}_\varepsilon^{\text{gc}}} = \exp \left\{ \mu_\varepsilon \langle (p, \psi), X_\varepsilon \rangle - \mu_\varepsilon Q_{\varepsilon, T}^{\text{gc}}(p, \psi) \right\}.$$

It is a mean-one density without introducing a contact compensator. Differentiability of the sewn pressure makes its mean converge to (f, Γ) ; the second derivative and exponential compactness give concentration under the tilt. The logarithmic density converges to the relative-entropy action by pointwise conjugacy. This proves the lower bound for regular pairs. Lemma 1.9 and lower semicontinuity extend it to every finite-action pair. $\square$

1.8. Microcanonical transfer. B1 supplies the exact finite-volume source-dependent saddle and a subexponential mixed shell coefficient for every bounded density/contact source.

Theorem 1.11 (Microcanonical joint LDP). *Under the primitive microcanonical shell, the same pair satisfies the good LDP with rate*

$$I_T^{\text{mc},a}(f, \Gamma) = I_{0,a}(f_0) + \int \ell \left(\frac{d\Gamma}{dA_f} \right) dA_f$$

on feasible absolutely continuous pairs and $+\infty$ otherwise. The zero-cost pair is the prepared Boltzmann solution and its actual limiting contact intensity.

Proof. Apply the B1 finite-volume saddle change of measure to every source used in the grand-canonical upper and lower bounds. The shell has subexponential probability under the exact tilted law, uniformly on bounded source balls. Only the initial rate changes from I_0^{gc} to $I_{0,a}$; the dynamic contact action and all compactness estimates are unchanged. At zero cost, $d\Gamma/dA_f = 1$ and the weak balance is the Boltzmann equation. $\square$

Theorem 1.12 (Round-six B2 closure). *Every actual deterministic hard-sphere contact is marked in a source-uniform fixed-horizon pressure theorem on a trace-regular hierarchy class. The first surplus contact is controlled by a full root-inclusive genealogical Gramian; all later contacts follow the unchanged true trajectory. The resulting grand-canonical and microcanonical density-actual-contact pairs satisfy the full good LDPs above.*

Proof. Combine Theorems 1.6, 1.7, 1.10, and 1.11, together with Proposition 1.1. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared local source, short-time, finite-horizon or Boltzmann–Grad regime specified in its theorems. Every load-bearing cross-paper input is typed in the round-three dependency ledger. Clean compilation and automated structural checks are necessary verification gates; independent external journal review remains a separate process.

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